*Tectonophysics*, 165 (1989) 29–39
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands



# Reorientation of remanent magnetism during tectonic fabric development: an example from the Waynesboro Formation, Pennsylvania, U.S.A.

JAMES R. VETTER Jr [1]*, KENNETH P. KODAMA [1] and ARTHUR GOLDSTEIN [2]

[1] *Department of Geological Sciences, Lehigh University, Bethlehem, PA 18015 (U.S.A.)*
[2] *Department of Geology, Colgate University, Hamilton, NY 13346 (U.S.A.)*

(Received January 1, 1988; revised version accepted November 17, 1988)

## Abstract

Vetter, Jr., J.R., Kodama, K.P. and Goldstein, A., 1989. Reorientation of remanent magnetism during tectonic fabric development: an example from the Waynesboro Formation, Pennsylvania, U.S.A. *Tectonophysics*, 165: 29–39.

A paleomagnetic study of the late Early–early Middle Cambrian Waynesboro Formation in south, central Pennsylvania reveals a pre-folding magnetization which is close to the expected Cambrian field at three sites, while other site mean directions are rotated by as much as 82° from the expected Cambrian field. These results are based on detailed thermal demagnetization analysis of 84 red siltstone and sandstone samples from nine sites collected over a 163 km$^2$ area. Local faulting, fault block rotation, and remanence acquisition during rapid apparent polar wander do not appear to be responsible for the observed remanence deviations.

Total strain at the sites, as indicated semiquantitatively by the anisotropy of magnetic susceptibility (AMS) of 28 samples, has been studied to test the hypothesis that deformation has rotated the remanence directions. AMS data vary considerably both between sites and within sites and show a path of progressive deformation on a Flinn diagram. There is a strong, positive correlation between the deviation of a sample's remanence from the expected Cambrian field and its position along the deformation path, suggesting that finite strain/fabric development has rotated the magnetizations.

The study area is structurally contained in the South Mountain Anticlinorium whose southern margin forms the displacement transfer zone to the southern Appalachian Alleghenian thrust sheets (Elliott, 1976). If thrusting had continued and incorporated the Waynesboro rocks into these thrust sheets, the rotated magnetizations could have been incorrectly interpreted to result from rigid thrust sheet rotation.

## Introduction

A growing number of paleomagnetists is concentrating their efforts on continental margins in an attempt to increase our understanding of the tectonics in orogenic belts (Van der Voo and Channell, 1980). Many of these studies use paleomagnetic declination to record thin-skinned rotations (e.g. Europe: Perroud, 1986; Balla, 1987; Marton and Elston, 1987; Asia: Klootwijk et al., 1986; and North America: Grubbs and Van der Voo, 1976; Watts and Van der Voo, 1979; Schwartz and Van der Voo, 1984).

However, a detailed look at those studies suggesting thrust sheet rotations in the Appalachians reveals some inconsistencies. From the Cambrian Rome Formation in Tennessee, French (1976) obtained pole positions which lay along 60° of arc

* Present address: Environmental Resources Management, Inc., Exton, Pa., U.S.A.

0040-1951/89/$03.50 © 1989 Elsevier Science Publishers B.V.

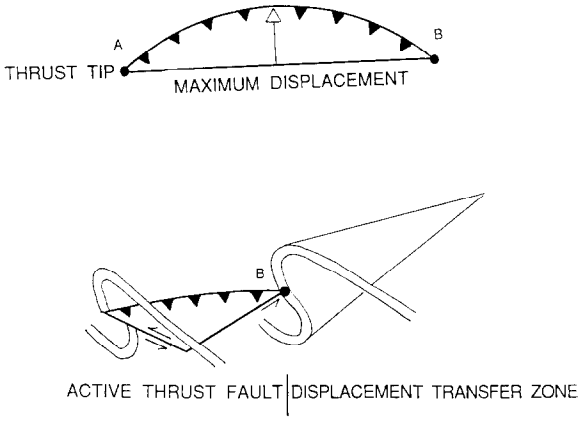



Fig. 1. Thrust fault propagation model after Elliot (1976). Thrust faults die out laterally at thrust tips which are adjacent to the unfaulted, asymmetrically folded rocks of the displacement transfer zone (dtz). Continued thrusting causes the tip to propagate laterally and incorporate these previously strained rocks.

and concluded that thrust sheet rotations could have given rise to the anomalous pole positions. Subsequent work by Watts et al. (1980a) on the Rome Formation and its northern stratigraphic equivalent, the Waynesboro Formation, ruled out remagnetization and nondipole field behavior as explanations and argued that tectonic rotations, the most plausible explanation, could not fully explain the results. They, therefore, concluded that rapid polar wander of North America during the Cambrian was responsible for the large swing in declination even though paleomagnetic studies (Elston and Bressler, 1977; Gillett and Van Alstine, 1979) suggest little apparent polar wander at this time.

Another possibility is that finite strain caused the observed deviations; therefore, to investigate this we have conducted a detailed paleomagnetic and anisotropy of magnetic susceptibility (AMS) study of the red sandstones and siltstones of the Waynesboro Formation in south-central Pennsylvania.

The Waynesboro Formation was selected because it is located in the displacement transfer zone (dtz) (Elliot, 1976) for the Alleghenian thrust sheets to the south. Elliot has postulated that thrust faults not only propagate forward but also laterally, extending their tips into the unfaulted asymmetrically folded strata outside the thrust sheet margins (Fig. 1). The folded strata directly adjacent to the thrust tip, which would have been incorporated into the thrust sheet if deformation had continued, are contained in the displacement transfer zone. Coward and Potts (1983) suggested that the short overturned limbs in the dtz experience combinations of high finite shear strain and layer-parallel shortening. Furthermore, folding and cleavage formation may add additional components of strain. Thus, rocks from many thrust sheets could have a pre-thrusting history of ductile strain and remanence rotation. It is possible, therefore, that remanence directions measured in such rocks could be mistakenly interpreted as recording thrust sheet rotations.

## Geologic history and setting

The Waynesboro Formation is estimated to be approximately 3100 m thick (Stose, 1906). A late Early Cambrian to early Middle Cambrian age has been assigned to the formation based on a sparse fauna which includes trilobites (Stose, 1909;

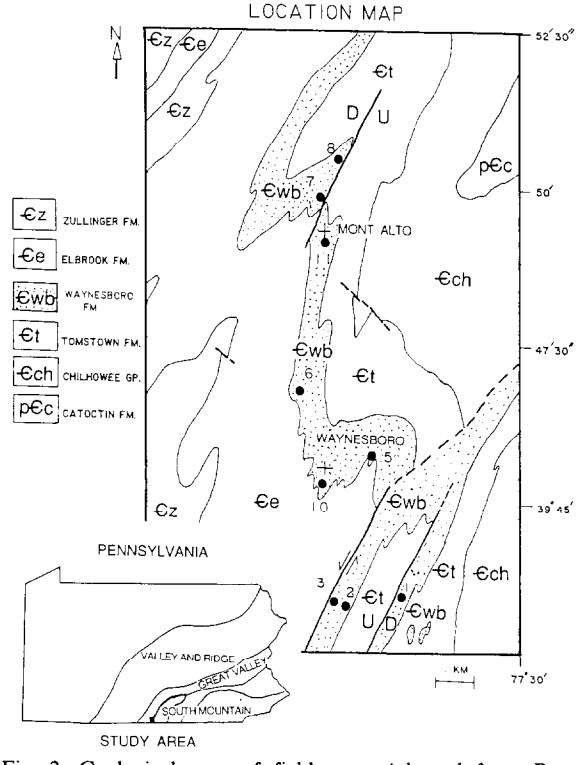



Fig. 2. Geological map of field area. Adapted from Root (1968).

Bassler, 1919). The formation consists of three members. The lowest is a buff, sandy dolomite. The stratigraphically thickest middle member is composed of dark blue limestone and dolomite. The upper member, which is the only unit sampled in this study, varies from sandy shale to siltstone and sandstone and is dark red to purple in color.

Detailed mapping by Root (1968) in southeastern Franklin County, Pennsylvania, shows that the geology of the study area is dominated by two major anticlines which plunge shallowly to the southwest and are cut by reverse and normal faults (Fig. 2). A regional, first generation, nearly vertical, axial planar cleavage strikes approximately N20° E. It is expressed in our study area as an intersection lineation on bedding surfaces, but is not obvious at the mesoscopic scale. Folds in the Waynesboro Formation have thickened hinges and thinned limbs suggesting that considerable plastic strain has accumulated.

## Sampling and measurement

Eighty-four oriented cores were collected at nine sites, distributed over a 163 $km^2$ area (Fig. 2). Site locations were constrained by the limited exposure of the upper member. Sites were chosen so that the effects of block rotation could be distinguished from the effects of grain scale deformation.

The mechanical incompetence of the upper member of the Waynesboro Fm. caused many of the cores to break during field sampling; consequently, only twenty-eight of the eighty-four 25.4 mm diameter cores could be trimmed to the optimum length (22.8 mm) for AMS determination. Bulk susceptibilities were measured on a Bison susceptibility meter. The orientation and ellipticity of the susceptibility ellipsoid were determined with a torque meter magnetometer. After AMS measurement, all cores were thermally demagnetized.

In a pilot demagnetization study, 16 cores from all nine sites were thermally demagnetized in 23 steps up to 700° C in a Schonstedt TSD-1 thermal demagnetizer. Remanence was measured with a Molspin Minispin spinner magnetometer. This machine was configured for high sensitivity and operated in a low field environment (less than 1000 nT) maintained by 2 m square Helmholtz coils. The noise level of the magnetometer was less than 0.02 mA/m and we have been able to repeat measurements at 0.08 mA/m. Based on the results from the demagnetization pilot study all remaining samples were thermally demagnetized with no fewer than 13 steps between 400° C and 700° C. Characteristic remanence (ChRM) directions were determined from least square lines fit to the demagnetization data using principal component analysis (Kirshvink, 1980). At least three demagnetization steps and the origin were used to calculate the ChRM.

## Results

The average intensity of natural remanent magnetization (NRM) for these samples was $3.42 \pm 3.08$ ($1\ \sigma$) mA/m with the NRM's having a mean direction of $I = 22.2°$, $D = 13.1°$ ($\alpha_{95} = 9.6°$), which is sufficiently close to the present field direction to suggest that a present day overprint is a significant portion of the NRM. The samples' bulk susceptibility ranged from $1.254 \times 10^{-4}$ to $2.50 \times 10^{-4}$ SI units, readily measured on the torque meter.

During thermal demagnetization most samples displayed a fairly simple, two component NRM (e.g. samples 6-1 and 7A-1; Fig. 3). A secondary, intermediate blocking temperature magnetization was removed at temperatures below 500° C. The vector endpoints on orthogonal plots (Zijderveld, 1967) decayed linearly into the origin for temperatures greater than 600° C. Most of the magnetization was removed over a short temperature interval between 625° C and 685° C (Fig. 3) suggesting that the main magnetic carrier was hematite or titanohematite with a limited range in grain size.

Thermal demagnetization of samples from site 2 did not result in isolation of a characteristic magnetization (Fig. 4) from any of the samples at the site; therefore, this site was removed from further consideration.

When the site mean directions were corrected for local bedding tilt (Fig. 5, Table 1) there was no statistical improvement in the amount of scatter. Application of stepwise unfolding at the sample

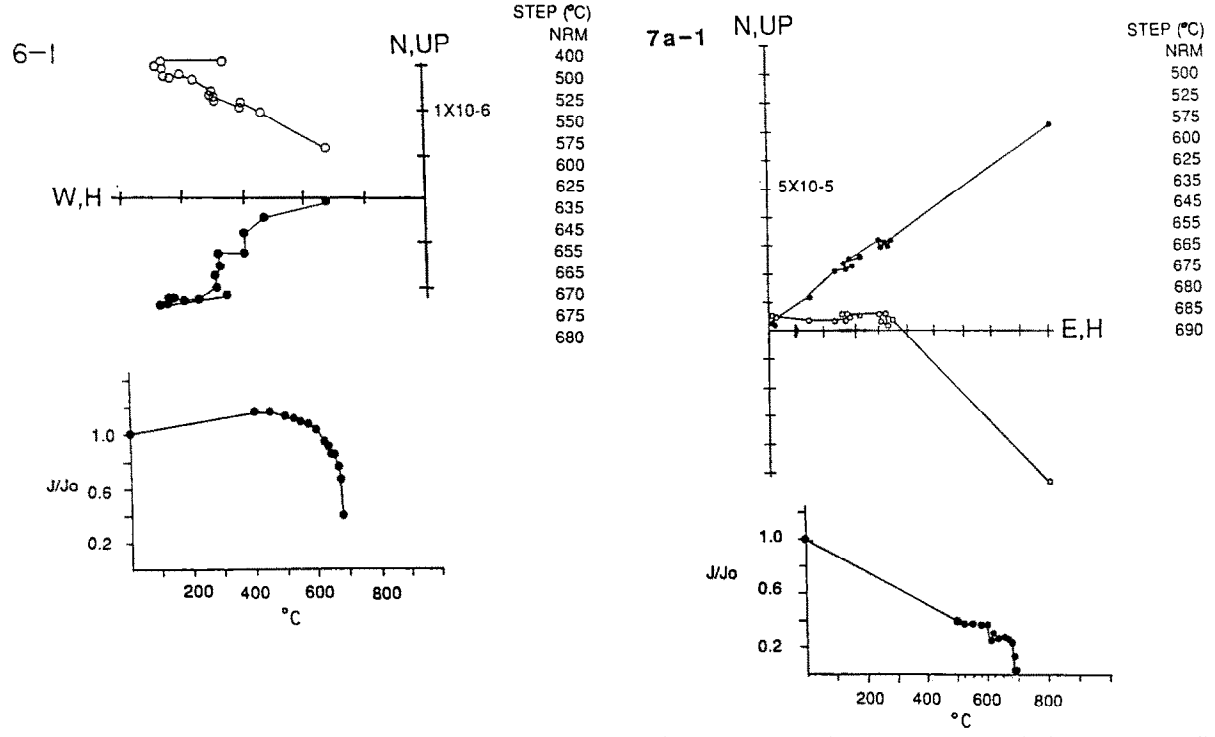



Fig. 3. Orthogonal projection and intensity plots for thermal demagnetization. Open circles represent the vertical component, solid circles represent the horizontal component. Both samples show a narrow unblocking temperature range and univectorial decay into the origin.

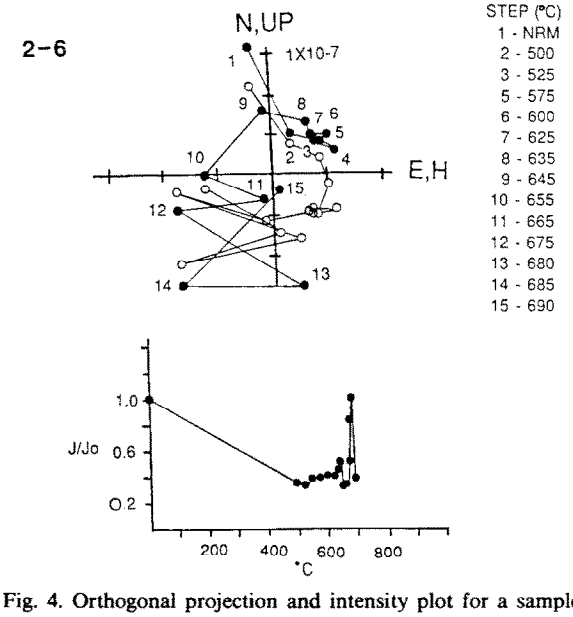



Fig. 4. Orthogonal projection and intensity plot for a sample representative of the behavior at site 2. Symbols the same as for Fig. 3. No stable direction was isolated during thermal treatment.

level ($N = 76$) indicated that the precision parameter, $K$ (Fisher, 1953), increased monotonically from 1.29 to 1.47, showing neither passage nor failure of the regional fold test. This contradicts the findings of Watts et al. ,1980a) who did find a statistically significant improvement. This difference may be due to our more detailed sampling of the Waynesboro Fm. (eight sites for this study, four sites for Watts et al., 1980a). However, since we will show that strain has affected the magnetization of these rocks, it is probably inappropriate to apply the fold test at a regional scale. Site 1, which is mildly strained (see Discussion) and has significant differences in the within site bedding tilts (some overturned beds based on sedimentary structures), passes the fold test at the 95% confidence level (Table 1) using the conservative estimates of McElhinny (1964). The mean direction for site 1 in stratigraphic coordinates is close to the field direction calculated from the Sept-Iles

TABLE 1

Site mean directions and statistics

| Site | $N$ | Geographic coordinates | | | | Stratigraphic Coordinates | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | dec. (°) | inc. (°) | K | $\alpha_{95}$ (°) | dec. (°) | inc. (°) | $K$ | $\alpha_{95}$ (°) |
| 1 | 7 | 158 | −7 | 1.9 | 38 | 155 | 40 | 6.5 | 20 |
| 3 | 7 | 165 | −3 | 2.9 | 30 | 195 | 49 | 3.1 | 29 |
| 5 | 6 | 062 | 21 | 5.0 | 25 | 053 | 25 | 4.2 | 27 |
| 6 | 7 | 098 | 45 | 2.3 | 34 | 341 | −77 | 2.3 | 34 |
| 7 | 10 | 237 | −0.4 | 1.2 | 39 | 236 | −6 | 1.2 | 38 |
| 8 | 8 | 135 | 66 | 3.8 | 25 | 132 | 45 | 3.5 | 26 |
| 10 | 19 | 124 | 12 | 4.3 | 15 | 128 | 21 | 4.3 | 15 |
| 11 | 12 | 292 | −3 | 2.6 | 24 | 290 | 24 | 2.5 | 25 |
| Mean of site means | | 150 | 33 | 1.5 | 30 | 168 | 49 | 1.3 | 42 |

remanence A pole (Tanczyk, et al., 1987). The age agreement between the 540 m.y. radiometric date for this pole and the fossil age of the Waynesboro Fm. is additional evidence that the remanence for site 1 is primary.

A weakly penetrative fabric evidenced by a consistently oriented cleavage-bedding intersection lineation was seen at all sites. Cleavage orientations, however, were not measurable at every site. In order to compare our AMS and remanence results to this incipient fabric, we calculated a reference cleavage plane (strike N21° E, dip 77° W; Fig. 5) which is the average cleavage orientation for the sites where cleavage was measurable. The regional consistency of cleavage orientation observed by Cloos (1947) and supported by our field observations make this procedure reasonable.

When the site mean directions are compared to the reference cleavage plane in geographic coordinates, two sites lie near the pole to the cleavage

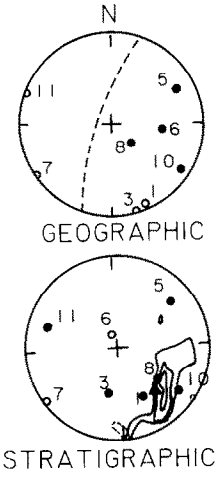



Fig. 5. Site mean directions in geographic and stratigraphic coordinates. Open circles represent upper hemisphere projections, solid circles represent lower hemisphere projections. A regional *S*1 cleavage is plotted by averaging cleavage measurements throughout the sampling area. Stratigraphic directions are plotted with respect to contours of the reversed polarity field directions calculated from the Cambrian poles tabulated by Tanczyk et al. (1987). Contour lines are 100%, 60%, and 20%. Solid lines are lower hemisphere; dashed lines are upper hemisphere. The triangle represents the field direction calculated from the Sept-Iles remanence A pole (Tanczyk et al., 1987).

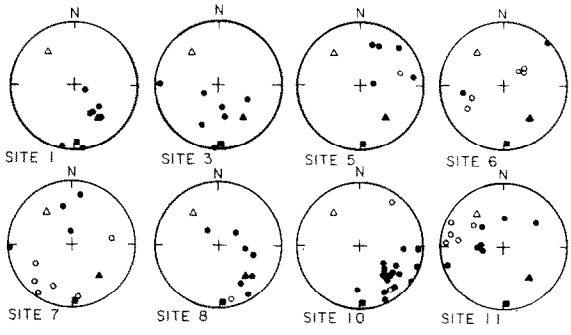



Fig. 6. Individual sample directions plotted by site for all samples collected from the Cambrian Waynesboro Formation. Triangles indicate the expected normal (northwesterly) and reversed (southeasterly) polarity field directions for the late Early/early Middle Cambrian calculated from the Sept-Iles remanence A pole (Tanczyk et al., 1987). Square represents the Kiaman (Permian) field for this site based on Irving and Irving's (1982) 240 m.y. pole for North America. Symbols as in Fig. 5.

plane (sites 10 and 11) and the rest weakly define a small circle about the pole to cleavage (Fig. 5). In stratigraphic coordinates the site means can be compared to the sweep of Cambrian directions tabulated by Tanczyk et al. (1987) and contoured in Fig. 5. Even though there is ambiguity in the expected Cambrian field direction due to the possibility of extensive polar wander, there is still an obvious scatter of site means away from the range in Cambrian directions, by as much as 90° in declination (sites 5 and 7). This behavior is seen to varying amounts within each site (Fig. 6) and in some cases results in a non-Fisherian distribution of the ChRM directions (sites 6, 7, 8, and 11; Fig. 6) at a site.

Although no reliable strain markers were observed in the field, thereby precluding a quantitative correlation between AMS and strain, our AMS results are a good indication of the relative strain magnitude between sites and between samples from an individual site. When plotted on a Flinn (1965) diagram (Fig. 7), the shapes of our susceptibility ellipsoids can be broken down into the five distinct domains which fall within the fields of apparent flattening and apparent constriction. These data suggest that the Waynesboro Formation has experienced varying degrees of strain distributed along a hypothetical strain path for the deformation of sediments. Burial will result in

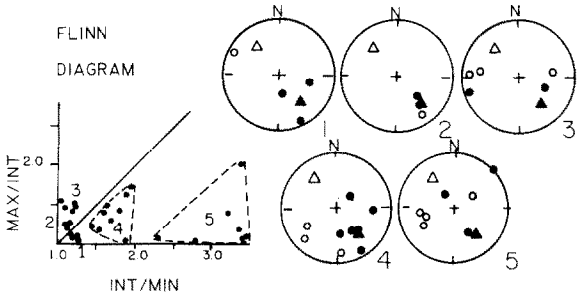



Fig. 7. Flinn (1965) diagram representing the shape of the three dimensional magnetic susceptibility ellipsoid for 28 samples from the Waynesboro Formation suggesting increasing strain from domain 1 through domain 5. Sample directions for each domain (indicated by the number below the stereonet) are plotted in stratigraphic coordinates and show a systematic rotation from the expected Cambrian field direction (represented by the triangles and calculated from the Sept-Iles remanence A pole—Tanczyk et al., 1987) as strain increases. Symbols for equal-area nets are the same as for Fig. 5.

flattened strain ellipsoids with subvertical short axes; horizontal compression results in shortening of one horizontal axis and extension in the vertical direction giving rise to prolate strain ellipsoids with sub-horizontal long axes; continued subhorizontal flattening leads to oblate ellipsoids with subhorizontal short axes (Ramsay and Huber, 1983; Kligfield et al., 1981). We have grouped our data into five domains which could represent various stages in this scenario with domain one representing the least strained and domain five the most strained samples. Samples within a single site do not cluster in one domain suggesting strain heterogeneity at the site level. Similar results have been noted by Kligfield et al. (1977) and Hirt et al. (1988).

The stereonets in Fig. 7 summarize the behavior of the remanence for individual samples in each of the five Flinn domains. Domains 1 and 2 have very mildly deformed samples and their ChRM's are tightly clustered near to the late Early/early Middle Cambrian field direction (represented by the Sept-Iles remanence A pole; Tanczyk et al., 1987) with an azimuthally symmetric distribution of directions. In domain 3 the ChRM's are more scattered about the reference direction. Since we apparently have two polarities present, there are two directional groups which are both rotated in a counterclockwise sense away from the late Early/early Middle Cambrian field direction. The ChRM's of domain 4 are centered around the expected Cambrian field direction and show a larger scatter than seen in domains 1 to 3. Domain 5 ChRM's show only one sample close to the late Early Cambrian/early Middle Cambrian direction. All the other samples show very large rotations from the Cambrian direction and lie in a vertical plane which strikes NE–SW. It is tempting to identify this as the regional cleavage plane but since these results are presented in stratigraphic coordinates each sample will have a different relationship to cleavage depending on its original bedding tilt. These data clearly demonstrate that with increasing strain and fabric development the remanence tends to be rotated away from the late Early/early Middle Cambrian field direction. Since the sense of rotation is counterclockwise, the remanence is even further from the Permian field

direction for this site ($I = 9.7°$, $D = 176.8°$, 240 m.y. pole, Irving and Irving, 1982).

## Discussion

Based on the fold test conducted at site 1, the magnetization of the Waynesboro Formation is pre-folding in age. Site 10, which is the same site as Watts et al.'s (1980a) site 1 for the Waynesboro Fm., showed no improvement in directional scatter after tilt correction but gave a mean direction close to our site 1. This result for site 10 is identical to that found by Watts et al. (1980a) and can probably be explained by the small bedding tilt (less than 20°) and our sampling of only the hinge and one limb of the fold due to limited outcrop exposure at the site. Field observations at both of these sites suggest that the rocks are relatively undeformed with cleavage barely discernible and intersection lineations, if seen, poorly defined. These results suggest that if the Waynesboro Fm. were unstrained throughout its area of exposure, it would carry a primary remanence at all sites which would give a paleomagnetic pole close to the Sept-Iles remanence A pole (Tanczyk et al., 1987). However, when site means are compared to the field direction calculated the Sept-Iles pole (Fig. 5), they are rotated from it by varying degrees.

Since the possibility of rapid apparent polar wander during the Cambrian (Watts et al., 1980b; Tanczyk et al., 1987) exists, some of the observed rotation of the site means could be due to an extended period of remanence acquisition by the Waynesboro Fm. during the Cambrian. Equally possible are varying degrees of Permo-Carboniferous reversed superchron (Kiaman) overprinting. However, a detailed look at the directions from each site (Fig. 6) suggests this is not the case. Some sites have directions consistent only with a late Early/early Middle Cambrian reference direction (sites 1, 8, 10) and some sites have directions which point northerly and steeply down

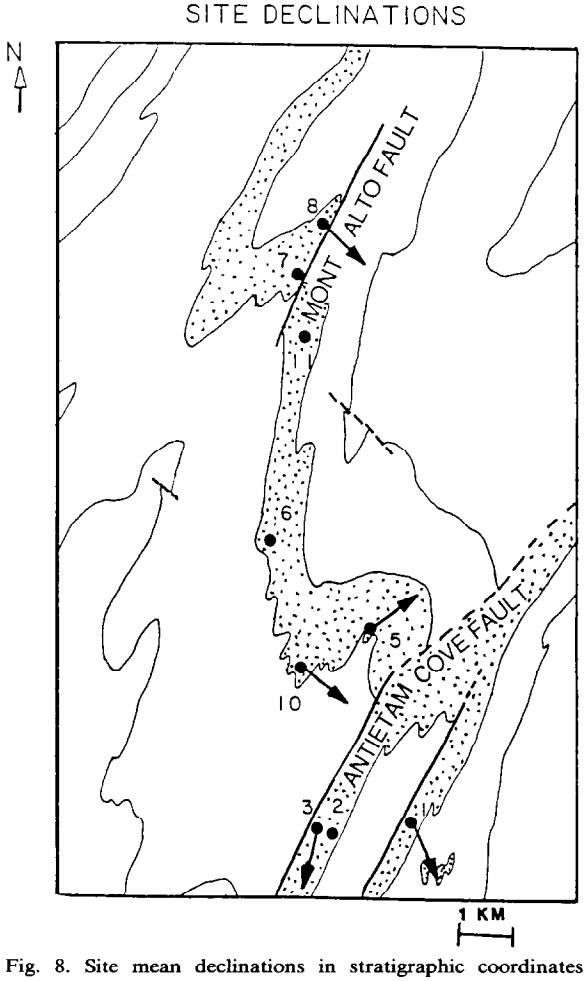



Fig. 8. Site mean declinations in stratigraphic coordinates plotted at individual sites. The arrows for sites 2, 6, 7, and 11 have been eliminated because of complications with the remanence at these sites (see text for discussion). A systematic rotation due to deformation near to faults or fault block rotation does not appear responsible for these deviations.

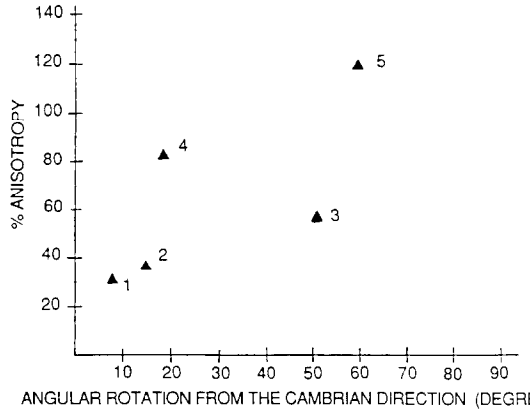



Fig. 9. Plot of rotation angle from the expected Cambrian field direction versus percent anisotropy for the mean directions from the Flinn domains in stratigraphic coordinates. Strain increases as the amount of vector rotation increases. Angles for the Flinn domains were calculated by assuming dual polarity and projecting normal polarity (positive inclinations) through the origin.

(sites 7, 8, 11) which may be due to unremoved present day/Brunhes chron field overprints. However, the remaining directions deviate from the Cambrian reference field and are far from the sweep in possible Cambrian directions or a Kiaman direction (sites 3, 5, 6, 7). The southwesterly/northeasterly directions from sites 5 and 7 could be consistent with poles 1 and 2 in Tanczyk et al.'s compilation (see contour representing the most northeasterly Cambrian field directions in Fig. 5); however, these poles are not based on modern day demagnetization techniques or analysis and are far from the main group of Cambrian poles.

Another explanation for the observed remanence deviations could be rigid fault block rota-

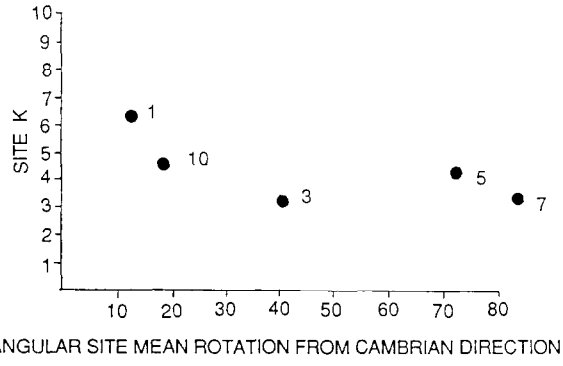



Fig. 10. Precision parameter $K$ versus great-circle angle from the expected Cambrian direction indicating increasing amounts of rotation are not accompanied by a dramatic increase in scatter. Normal polarity directions from site 7 were projected through the origin for this calculation.

TABLE 2

AMS results

| Sample | Anisotropy (%) | AMS principle axis ratios | | | Flinn graph domain |
|---|---|---|---|---|---|
| | | $K_{max}/K_{int}$ | $K_{int}/K_{min}$ | $K_{max}/K_{min}$ | |
| 1-1 | 32 | 1.21 | 1.13 | 1.38 | 2 |
| 1-3 | 53 | 1.19 | 1.45 | 1.74 | 4 |
| 1-5 | 35 | 1.27 | 1.11 | 1.42 | 2 |
| 1-6 | 171 | 2.09 | 4.82 | 10.11 | 5 |
| 3-1 | 64 | 1.49 | 1.26 | 1.89 | 3 |
| 3-5 | 80 | 1.42 | 1.63 | 2.32 | 4 |
| 3-7 | 31 | 1.06 | 1.30 | 1.39 | 1 |
| 6-1 | 141 | 1.10 | 12.96 | 14.35 | 5 |
| 6-3 | 130 | 1.01 | 10.41 | 10.61 | 5 |
| 6-5 | 110 | 1.21 | 3.43 | 4.16 | 5 |
| 6-6 | 126 | 1.48 | 3.42 | 4.82 | 5 |
| 6-7 | 86 | 1.04 | 2.81 | 2.94 | 5 |
| 7A-1 | 54 | 1.58 | 1.06 | 1.67 | 3 |
| 7A-3 | 87 | 1.50 | 1.67 | 2.51 | 4 |
| 7B-4 | 89 | 1.44 | 1.83 | 2.64 | 4 |
| 7B-5 | 59 | 1.19 | 1.58 | 1.88 | 4 |
| 8-1 | 29 | 1.08 | 1.25 | 1.35 | 1 |
| 8-4 | 79 | 1.08 | 2.35 | 2.57 | 5 |
| 8-6 | 61 | 1.04 | 1.91 | 2.01 | 4 |
| 8-7 | 107 | 1.65 | 1.91 | 3.17 | 4 |
| 10-3 | 115 | 1.74 | 1.99 | 3.48 | 4 |
| 10-8a | 77 | 1.31 | 1.72 | 2.27 | 4 |
| 10-14 | 45 | 1.30 | 1.20 | 1.56 | 2 |
| 10-18 | 32 | 1.13 | 1.22 | 1.38 | 1 |
| 11-1 | 34 | 1.11 | 1.28 | 1.42 | 1 |
| 11-2 | 64 | 1.50 | 1.25 | 1.89 | 3 |
| 11-4 | 53 | 1.48 | 1.12 | 1.68 | 3 |
| 11-6 | 57 | 1.44 | 1.22 | 1.76 | 3 |

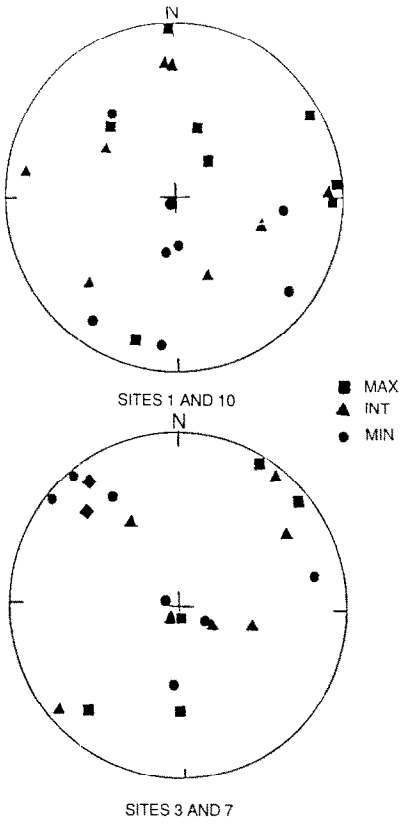



Fig. 11. Principal AMS axis directions in geographic coordinates for unstrained sites 1 and 10 and strained sites 3 and 7 indicating a preferred grain orientation at higher strains.

tions or rock deformation caused by local fault activity. Sites 5, 6, and 10 are located near the center of a large fault block (Fig. 2). If rigid block rotations had occurred, these vectors should have a similar sense of rotation. Site 6 cannot be used due to it scattered directions. However, sites 5 and 10 have very different declinations (Fig. 8) suggesting that a rigid block rotation cannot explain the remanence rotations. Rock deformation effects near to faults are ruled out by the observation that sites 1 and 8 have directions nearly parallel to the late Early/early Middle Cambrian direction.

Since large-scale rigid block rotations, faulting effects, and remanence acquisition during rapid apw fail to adequately explain the paleomagnetic vector rotations in the Waynesboro Fm., we suggest that rock deformation and fabric development at a small scale is the cause. Examination of the Flinn (1965) graph (Fig. 7) indicates that as a sample moves along the deformation path, its remanence is rotated from the late Early/early Middle Cambrian field direction and its magnetic susceptibility anisotropy increases (Fig. 9). The suggests that increased strain, quantified by AMS, has affected the remanence of the Waynesboro Formation. However, the samples which make up any one Flinn domain come from different sites so any given site may contain samples from several Flinn domains. This points to the small scale at which the rock deformation/fabric development affects the remanence and would suggest that a non-Fisherian distribution of directions and very large directional scatter would result at sites which have experienced extensive deformation (Kligfield et al., 1983).

TABLE 3

Orientation of principle axes of AMS ellipsoids

| Sample | $K_{max}$ | | $K_{intermediate}$ | | $K_{min}$ | |
|---|---|---|---|---|---|---|
| | $D$ (°) | $I$ (°) | $D$ (°) | $I$ (°) | $D$ (°) | $I$ (°) |
| 1-1 | 60 | 8 | 159 | 49 | 323 | 40 |
| 1-3 | 197 | 14 | 305 | 50 | 97 | 36 |
| 1-5 | 46 | 70 | 279 | 12 | 187 | 15 |
| 1-6 | 18 | 56 | 227 | 30 | 129 | 14 |
| 3-1 | 182 | 38 | 328 | 46 | 78 | 18 |
| 3-5 | 314 | 24 | 57 | 26 | 187 | 52 |
| 3-7 | 50 | 8 | 223 | 82 | 320 | 1 |
| 6-1 | 40 | 18 | 147 | 44 | 294 | 41 |
| 6-3 | 54 | 28 | 173 | 43 | 304 | 34 |
| 6-5 | 185 | 31 | 69 | 35 | 304 | 38 |
| 6-6 | 11 | 36 | 261 | 25 | 145 | 43 |
| 6-7 | 31 | 14 | 145 | 58 | 294 | 28 |
| 7-1a | 227 | 20 | 107 | 53 | 329 | 29 |
| 7-3a | 182 | 85 | 38 | 4 | 308 | 3 |
| 7-4b | 323 | 13 | 232 | 5 | 121 | 76 |
| 7-5b | 32 | 2 | 122 | 72 | 288 | 83 |
| 8-1 | 105 | 45 | 270 | 44 | 8 | 7 |
| 8-4 | 26 | 1 | 116 | 22 | 295 | 68 |
| 8-6 | 48 | 7 | 142 | 24 | 301 | 65 |
| 8-7 | 124 | 4 | 225 | 71 | 33 | 19 |
| 10-3 | 317 | 42 | 108 | 45 | 213 | 15 |
| 10-8a | 358 | 3 | 89 | 9 | 252 | 87 |
| 10-14 | 86 | 1 | 356 | 24 | 177 | 66 |
| 10-18 | 92 | 5 | 359 | 26 | 192 | 63 |
| 11-1 | 137 | 18 | 47 | 2 | 310 | 72 |
| 11-2 | 145 | 54 | 314 | 36 | 48 | 5 |
| 11-4 | 151 | 51 | 312 | 37 | 49 | 9 |
| 11-6 | 165 | 50 | 299 | 30 | 43 | 24 |

A detailed look at each site's directional scatter indicates that the postulated increase in scatter is not always seen (Fig. 6). When the estimate of Fisher's (1953) precision parameter $K$ is plotted as a function of the site mean's rotation from the expected Cambrian field direction for sites 1, 3, 5, 7, and 10, there is no precipitous drop in $K$ for rotations as great as 82° (Fig. 10, Table 1). In addition, the distributions of ChRM's at sites 3 and 5 (Fig. 6) appear sufficiently azimuthally symmetric so that one may not suspect that strain has deformed the remanence, yet their site means have been rotated by considerable amounts from the late Early/early Middle Cambrian field direction.

A comparison of the AMS results from rotated sites 3 and 7 and unrotated sites 1 and 10 shows that the degree of fabric development controls the remanence rotation. The principal AMS axes of seven samples from sites 3 and 7 show that a preferred grain orientation is developing in these samples (Fig. 11, Table 3) with the $K_{min}$ axes clustering perpendicularly to the regional cleavage plane. The AMS of eight samples from unrotated sites 1 and 10 show that even though some of these samples may have experienced large strains, fabric development has not proceeded far enough to cause rotation of the remanence at these sites.

## Conclusions

Although the red sandstones and siltstones of the upper member of the Cambrian Waynesboro formation appear, from field observations, to be weakly deformed and could have been considered suitable for paleomagnetic study (e.g., Watts et al., 1980a) these rocks carry a primary Cambrian remanence that has been rotated by varying degrees from the expected late Early/early Middle Cambrian field direction. When AMS data are displayed on a Flinn diagram, they suggest the rocks have undergone different amounts of tectonic strain following compaction. The location of a site's mean remanence direction can be correlated to the average position of its samples on the Flinn graph's deformation path and to the degree to which the magnetic grains of the samples show a preferred alignment as indicated by the scatter of the principal AMS axes.

The Waynesboro Formation lies within the displacement transfer zone (Elliot, 1976) of the Alleghenian thrust sheets to the south. Had thin-skinned thrusting continued during the Alleghenian orogeny, the strain-rotated remanence of the Waynesboro Fm. would have been incorporated into the Alleghenian thrust sheets and could have been misinterpreted as indicating rigid thrust sheet rotation. Our study shows that finite strain associated with tectonic fabric development must always be investigated as one possible explanation for rotated remanence when paleomagnetism is used to study tectonic rotations in orogenic belts.

## Acknowledgments

This study was funded, in part, through the generous contributions of the Standard Oil Company of California and Sigma Xi and NSF grant EAR-8417312.

## References

Balla, Z., 1987. Tertiary palaeomagnetic data for the Carpatho-Pannonian region in light of Miocene rotation kinematics, Tectonophysics, 139: 67–98.

Bassler, R.H., 1919. Cambrian and Ordovician Deposits of Maryland. Maryland Geological Survey—John Hopkins Press, Baltimore, Md, 242 pp.

Cloos, E., 1947. Oolite deformation in the South Mountain Fold. Bull. Geol. Soc. Am., 58: 843–918.

Coward, M.P. and Potts, G.S., 1983. Complex strain patterns developed at the frontal and lateral tips to shear zones and thrust faults. J. Struct. Geol., 5: 383–399.

Elliott, P., 1976. The energy balance and deformation mechanisms of thrust faults. Philos. Trans. R. Soc. London, 283: 289–312.

Elston, D.P. and Bressler, D.R., 1977. Paleomagnetism of Lower Cambrian sedimentary rocks from Devonian strata of Arizona. Earth Planet. Sci. Lett., 36: 423–433.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc., London, Ser. A, 217: 295–305.

Flinn, D., 1965. On the symmetry principle and the deformation ellipsoid. Geol. Mag., 102: 36–45.

French, R.B., 1976. Lower Paleozoic Paleomagnetism of the North American Craton. PhD Thesis, University of Michigan, Ann Arbor, 150 pp.

Gillett, S.C. and Van Alstine, D.R., 1979. Paleomagnetism of Lower and Middle Cambrian sedimentary rocks of the Desert Range, Nevada. J. Geophys. Res., 84: 4475–4489.

Grubbs, K.L. and Van der Voo, R., 1976. Structural deforma-

tion of the Idaho–Wyoming Overthrust Belt as determined by Triassic paleomagnetism. Tectonophysics, 33: 321–336.

Hirt, A.M., Lowrie, W., Clendenen, W.S. and Kligfield, R., 1988. The correlation of magnetic anisotropy with strain in the Chelmsford Formation of the Sudbury Basin, Ontario. Tectonophysics, 145: 177–189.

Irving, E. and Irving, G.A., 1982. Apparent polar wander paths Carboniferous through Cenozoic and the assembly of Gondwana. Geophys. Surv., 5: 141–188.

Kirshvink, J.S., 1980. The least squares line and plane analysis of paleomagnetic data. Geophys. J.R. Astron. Soc., 62: 699–718.

Kligfield, R., Lowrie, W. and Dalziel, I.W.D., 1977. Magnetic susceptibility as a strain indicator in the Sudbury Basin, Ontario. Tectonophysics, 40: 287–308.

Kligfield, R., Owens, W.H. and Lowrie, W., 1981. Magnetic susceptibility anisotropy, strain and progressive strain in permian sediments of the Maritime Alps (France). Earth Planet. Sci. Lett., 55: 181–189.

Kligfield, R., Lowrie, W. and Siddan, A.W.B., 1983. Effect of progressive deformation on remanent magnetization of Permian redbeds from the Alpes Maritimes (France). Tectonophysics, 97: 59–85.

Klootwijk, C., Sharma, M., Gergan, J., Shah, S. and Gupta, B., 1986. Rotational overthrusting of the northwestern Himalaya: further palaeomagnetic evidence from the Riasi thrust sheet, Jammu foothills, India. Earth Planet. Sci. Lett., 80: 375–393.

Marton, E. and Elston, D.P., 1987. Tectonic rotations south of the Bohemian Massif from palaeomagnetic directions of Permian red beds in Hungary. Tectonophysics, 139: 43–51.

McElhinny, M.W., 1964. Statistical significance of the fold test in paleomagnetism. Geophys. Trans. R. Astron. Soc, 8: 338–348.

Perroud, H., 1986. Paleomagnetic evidence for tectonic rotations in the Variscan Mountain belt. Tectonics, 5: 205–214.

Ramsey, J.G. and Huber, M.I., 1983. Techniques of Modern Structural Geology. Vol. 1. Strain Analysis. Academic Press, London, 307 pp.

Root, S.I., 1968. Geology and mineral resources of southeastern Franklin County, Pennsylvania. Pa. Geol. Survey Atlas, 119c: 93 pp.

Schwartz, S.K. and Van der Voo, R., 1984. Paleomagnetic study of thrust sheet rotation during foreland impingement in the Wyoming–Idaho Overthrust belt. J. Geophys. Res., 89: 10077–10086.

Stose, G.W., 1906. The sedimentary rocks of South Mountain, Pennsylvania. J. Geol., 14: 201–220.

Stose, G.W., 1909. Description of the Mercersburg–Chambersburg districts Pennsylvania, U.S. Geol. Surv., Survey Atlas, Folio 170, 19 pp.

Tanczyk, E.I., LaPointe, P., Morris, W.A. and Schmidt, P.W., 1987. A paleomagnetic study of the layered intrusion at Sept-Iles Quebec. Can. J. Earth Sci., 24: 1431–1438.

Van der Voo, R. and Channell J.E.T., 1980. Paleomagnetism in orogenic belts. Rev. Geophys. Space Phys., 18 (2): 455–481.

Watts, D.R. and Van der Voo, R., 1979. Paleomagnetic results from the Ordovician Moccasin Bays and Chapman Ridge formations of the Valley and Ridge province, eastern Tennessee. J. Geophys. Res., 85: 645–655.

Watts, D.R., Van der Voo, R. and French, R.B., 1980a. Paleomagnetic study of the Cambrian Waynesboro and Rome formations of the Valley and Ridge province of the Appalachian Mountains. J. Geophys. Res., 84: 5331–5394.

Watts, D.R., Van der Voo, R. and Reeve, S.C., 1980b. Cambrian paleomagnetism of the Llano Uplift, Texas. J. Geophys. Res., 85: 5316–5330.

Zijderveld, J.D.A., 1976. Demagnetization of rocks: analysis of results. In: D.W. Collinson, K.M. Creer and K. Runcorn (Editors), Methods in Paleomagnetism. Elsevier, Amsterdam, pp. 254–286.